Saman Nezafat

Github: github.com/onionj | **Email**: samannzft@gmail.com | **LinkedIn**: www.linkedin.com/in/samannzft

Professional Summary

Backend Software Engineer with 5 years of experience architecting high-concurrency systems and real-time platforms serving millions of users. Proven track record in technical leadership and delivering scalable, distributed solutions in Go and Python. Open to relocation with visa sponsorship.

Work Experience

Intelligent Processing of Pars Hadish - 2023-Present

- Stack: Go, Python, FastAPI, PostgreSQL, MongoDB, ClickHouse, Redis, RabbitMQ, WebSocket, Docker

B2C Tourism Platform

- Tech-led a **tourism platform (3M+ users, 9+ B2B partners)** with a 2-engineer team, covering payment routing to per-account settlement terminals and a reconciliation system of 3 interacting state machines for automated and manual mismatch resolution.
- Built corporate benefit-card contracts (quota- or balance-based entitlements per employee) and a multi-case **refund engine** (discount codes, disability subsidies, corporate credits); redesigned DB connection lifecycle and **Redis caching** to sustain **50,000 concurrent peak users**.

Payment Gateway

- Built and maintain the original production payment layer in **Python/FastAPI** for the tourism platform (5 integrated gateways).
- Built a **Go** rewrite with a cleaner provider-adapter architecture (Make/Verify/Reverse/Refund per provider), **Redis distributed locking**, and **idempotency** guarantees; now used by two Shiraz municipal projects.

Audit Log Service

- Built a tamper-evident audit log service in **Go** with **per-source HMAC-SHA256 hash chains** — each connected backend has its own independent chain — a **Redis Streams** → **ClickHouse** high-throughput ingestion pipeline, and an emergency mode with disk monitoring and automatic log rotation.

Nekisa — Building Management & Security Monitoring

- Contributed to a **building management and security platform** running **4,500+ IoT devices** (cameras, access control, barriers) in production at Atlas Mall Tehran and Shiraz Metro, where **RabbitMQ** device streams drive a configurable scenario engine — surfacing a zone's cameras the moment a door opens.
- Integrated several new device types into the platform, including **camera tamper detection** that stores the shock-wave signal and a snapshot when a unit is physically disturbed; now hardening the platform toward a formal security certification.

Meter Data Management System

- Developed a **Go** backend that collects meter data and runs a multi-tier billing engine to automatically generate detailed PDF invoices.

Real-Time User Analytics System

- Built a high-throughput analytics system with a visual **Journey Builder** to automate personalized user flows, engagement triggers, and A/B tests; powered by **FastAPI**, **ClickHouse**, and **RabbitMQ**.

Messaging System

- Built a priority-managed messaging system (SMS & Push) using **Redis Streams** for efficient, real-time message queuing.

Real-Time Taxi POS Backend

- Developed a **WebSocket-powered** system for real-time **POS payments** and **live location streaming**.

AI-Powered Supermarket Agent

- Built an automated shopping assistant for WhatsApp, enabling users to place orders via voice or text, with the AI handling product search, cart management, and order creation.

Atishahr Smart City Iranians (Atishahr) - 2022-2023

- Stack: Python, FastAPI, PostgreSQL

SIB Ticket System

- Optimized a key reporting endpoint, achieving a **20x performance** improvement and eliminating slowdowns under high load.
- Contributed to the project's second version using **FastAPI**, implementing **asynchronous features** to handle increased system load.

Mehr-e Pars ICT (Mehr Pars) - 2021-2022

- Stack: Python, FastAPI, PostgreSQL

Organizational Resource Management System

- Engineered a **data migration** solution, enabling seamless transitions from legacy project management systems to the Rainesh platform.

Comprehensive Warehouse System

- Developed a **load-simulating reverse proxy** module for a large-scale warehouse system, simulating backend conditions under high load to improve frontend stability.

Open Source

- **Muxr**, A Go WebSocket multiplexing library for efficient real-time communications.
- **IP**, A Go TCP server that returns the client's IP and country in multiple formats.
- **PriceBot**, A Go Telegram bot providing real-time currency exchange rates, gold prices, and cryptocurrency values.
- **Social Token Experiment**, A minimalist ERC-20 social token on Polygon, designed as an experiment in on-chain communication (Solidity).
- **PyBotNet**, An educational security-research framework in Python for exploring remote-control patterns via Telegram.

Technical Skills

- **Languages**: Go, Python, Solidity
- **Frameworks & Protocols**: FastAPI, Gin, REST, WebSocket
- **Databases & Caching**: PostgreSQL, MongoDB, ClickHouse, Redis
- **Messaging & Streaming**: RabbitMQ, Redis Streams
- **Architecture**: Distributed systems, microservices, event-driven design, high concurrency, idempotency
- **DevOps**: Linux, Docker, Git, CI/CD
- **Security**: JWT, OWASP best practices

Soft Skills

- **Leadership & Collaboration**: Led a 3-member engineering team, guiding system design and code reviews while cultivating collaboration through mentoring and documentation.

Education

- **Bachelor of Science in Civil Engineering**

Languages

- **English**: Professional Working Proficiency